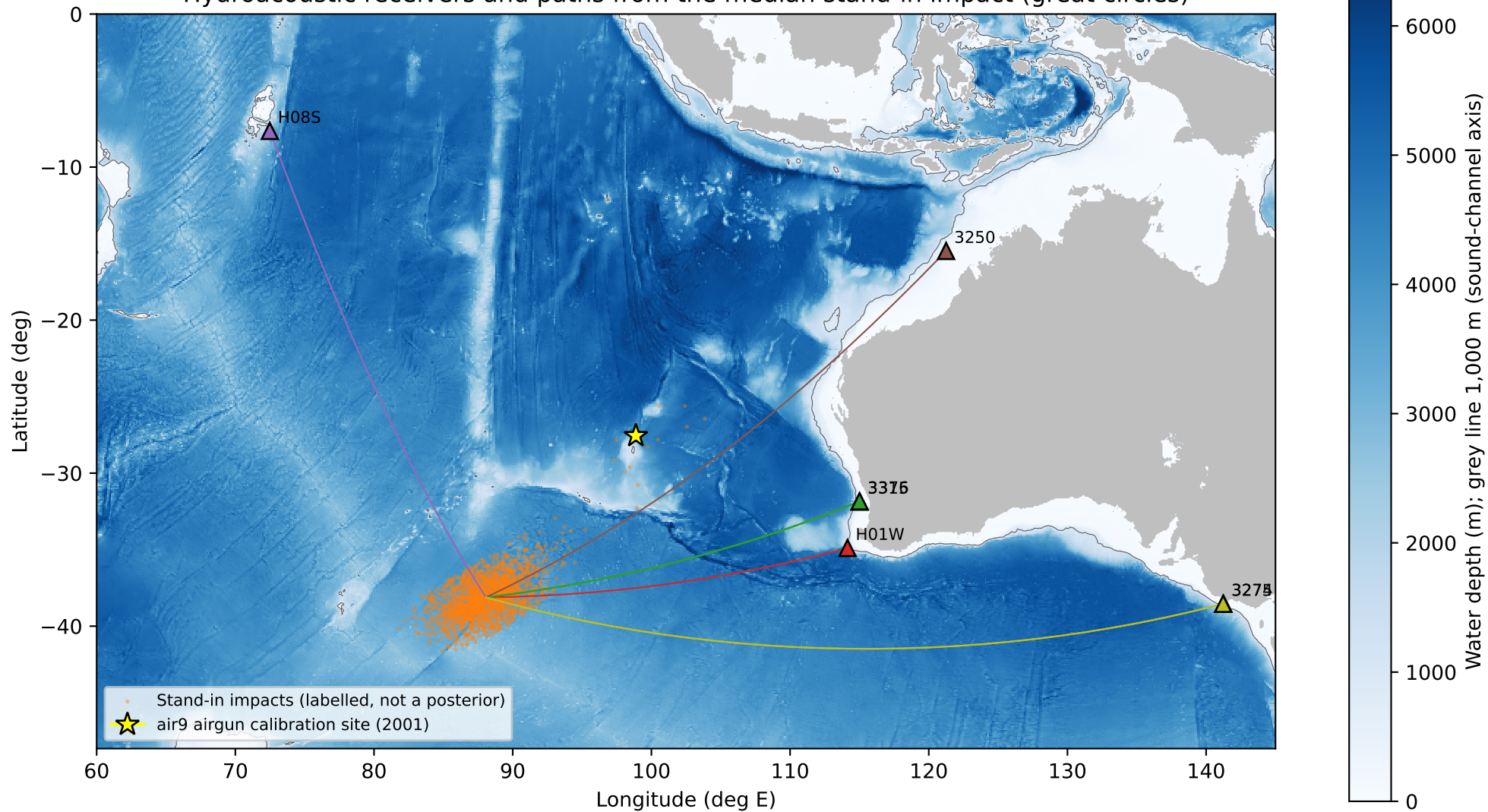
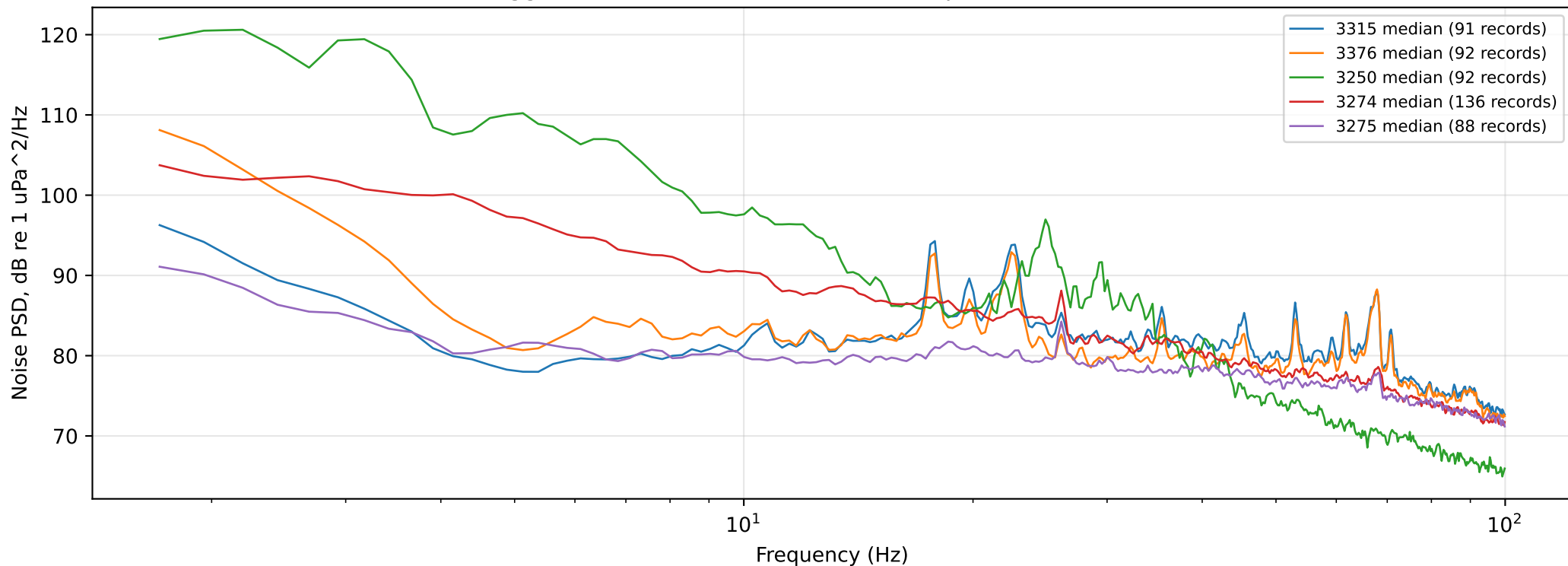


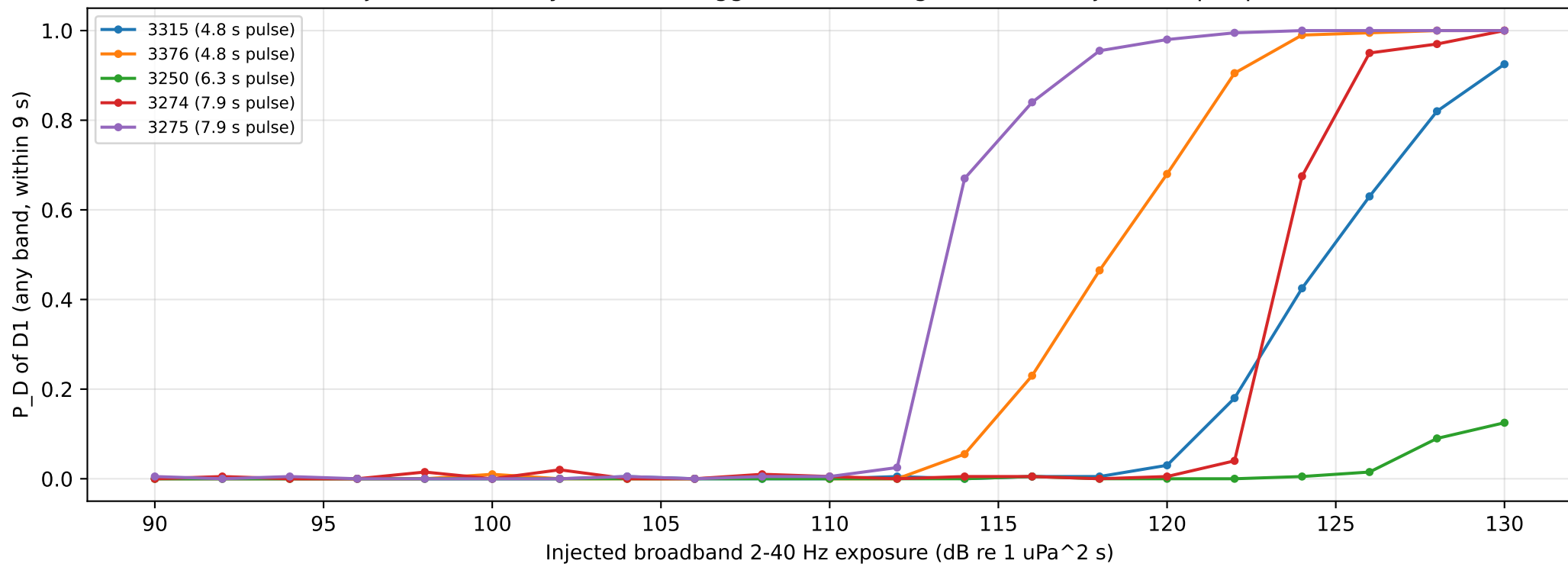
Hydroacoustic receivers and paths from the median stand-in impact (great circles)



IMOS loggers, 8-9 March 2014 outside the quarantine: median noise



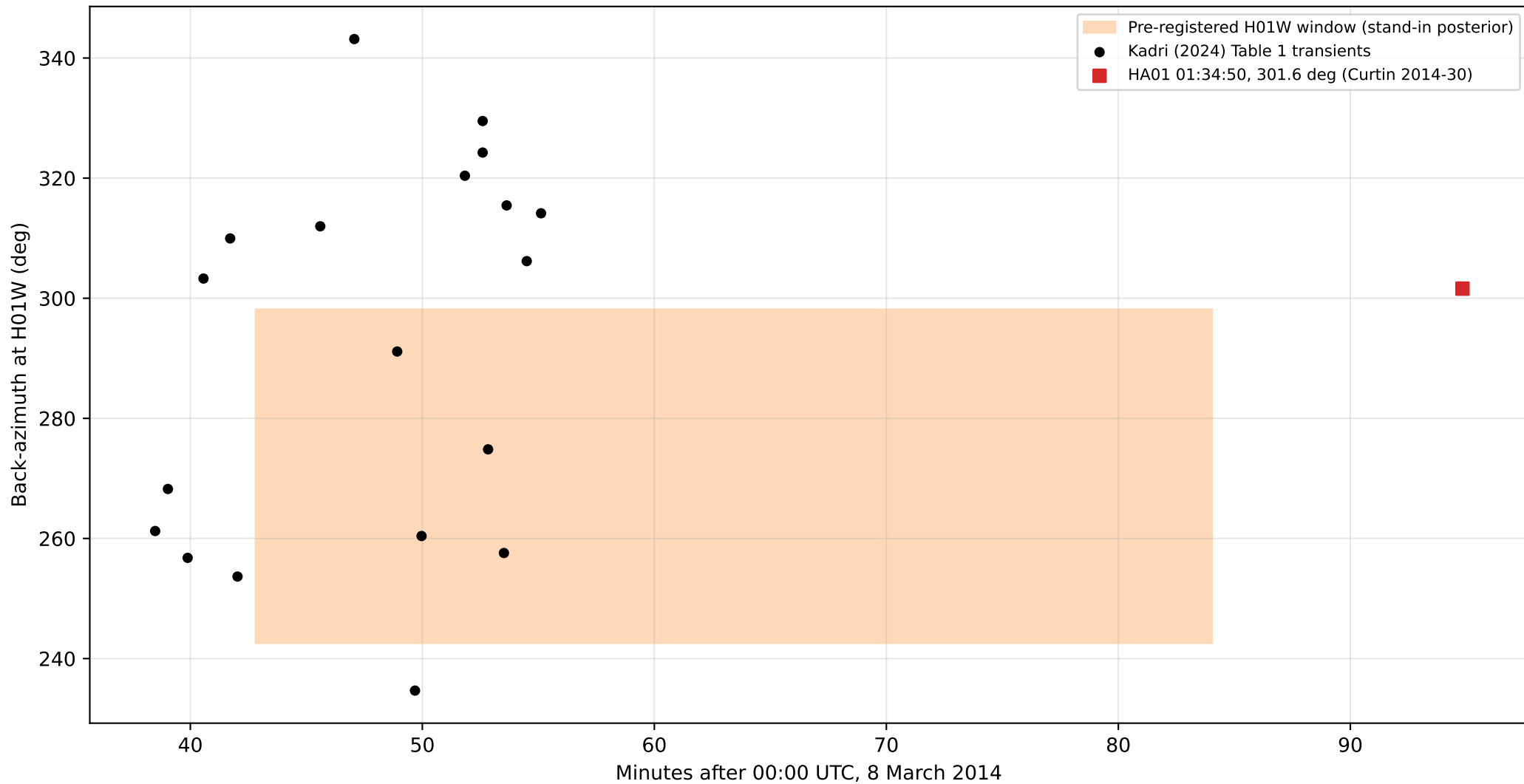
Injection-recovery into each logger's own background (200 injections per point)



Detection control, IMOS loggers (8-9 March 2014, outside the quarantine)

logger	hours	thresholds S (dB)	2-5/5-10/10-20/20-40	exposure for P_D = 0.5
3315	7.4	25.2/31.6/21.6/18.2		125 dB re 1 uPa^2 s
	control 05:03:01 UTC: not found; 2-5 Hz S=1.2 (159.72/h as strong), 5-10 Hz S=10.2 (2.02/h as strong), 10-20 Hz S=14.8 (1.89/h as strong), 20-40 Hz S=17.3 (0.3/h as strong)			
	control 05:18:13 UTC: not found; 2-5 Hz S=1.4 (146.37/h as strong), 5-10 Hz S=3.6 (27.11/h as strong), 10-20 Hz S=5.2 (29.54/h as strong), 20-40 Hz S=8.9 (9.3/h as strong)			
	screened out: 531B2CDD.DAT (30.08 s of identical A/D counts)			
3376	7.5	5.6/13.4/34.2/25.4		118 dB re 1 uPa^2 s
3250	7.5	33.0/44.0/39.1/33.7		> 130 dB (not reached)
3274	11.1	17.6/19.5/19.4/17.9		123 dB re 1 uPa^2 s
3275	7.2	23.9/16.3/18.6/12.0		113 dB re 1 uPa^2 s

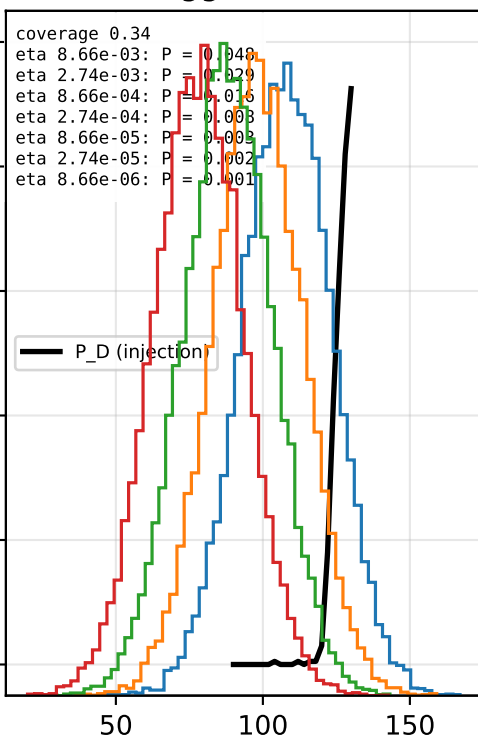
H01W arrivals against the pre-registered time and bearing window



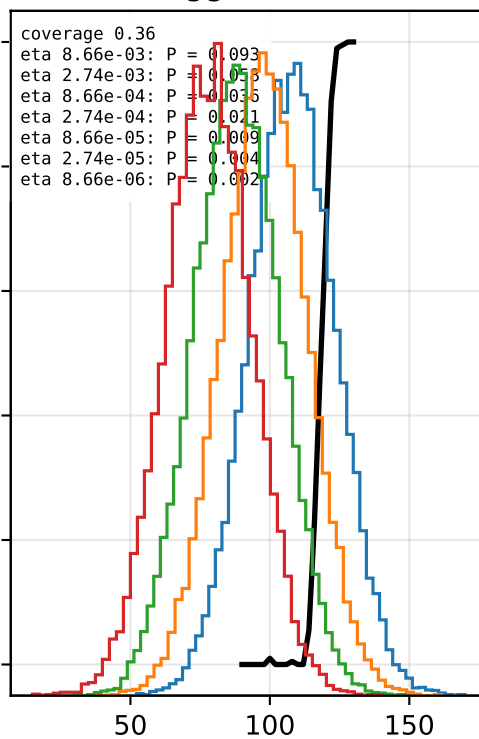
Predicted exposure over the labelled stand-in impacts (nominal 4 GJ; TL incl. 15 +/- 10 dB site coupling) against measured P_D. P = mean over impacts of record coverage x P_D, zero if blocked

P_D (black); predicted exposure density (coloured, own scale)

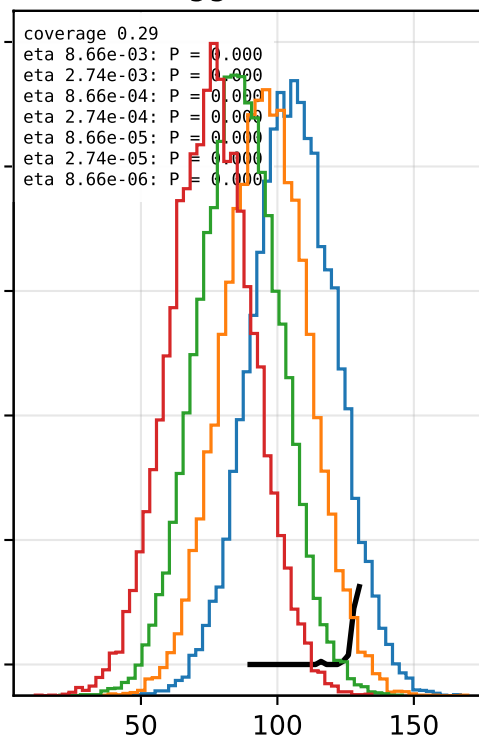
Logger 3315



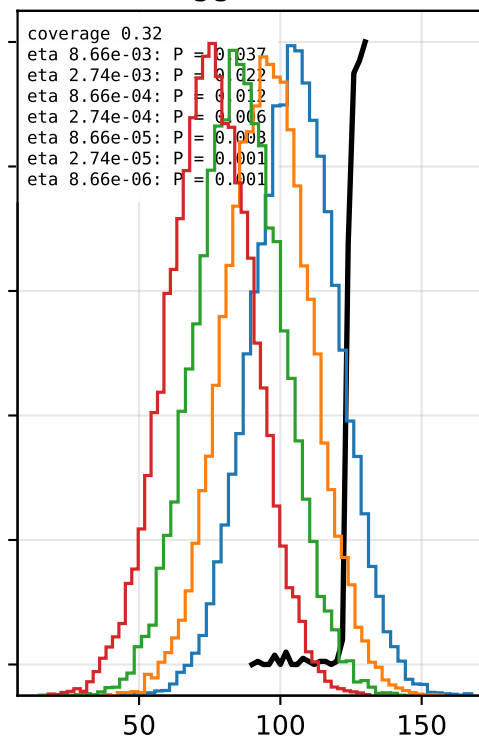
Logger 3376



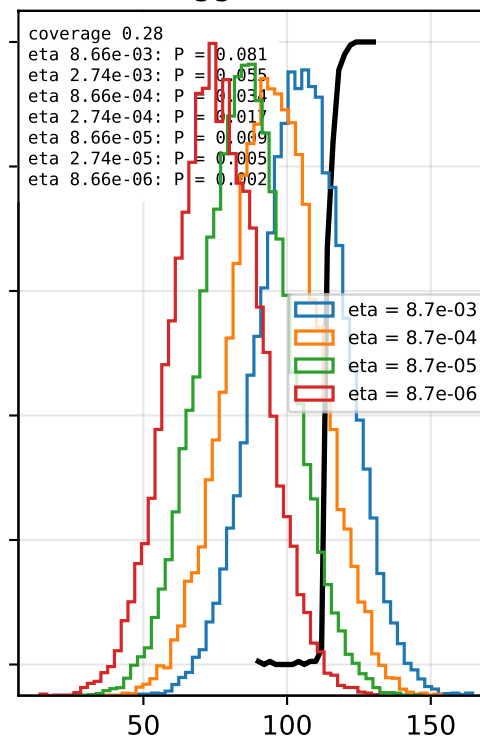
Logger 3250



Logger 3274



Logger 3275



Broadband exposure (dB re 1 uPa² s) Broadband exposure (dB re 1 uPa² s) Broadband exposure (dB re 1 uPa² s) Broadband exposure (dB re 1 uPa² s) Broadband exposure (dB re 1 uPa² s)

Pre-registered search of the IMOS loggers (windows from the stand-in envelope; thresholds fixed from background)

logger	window (UTC)	searched	expected	found	P(N>=n)
3315	00:44:31-01:26:17	880s	0.20	0	1.00
3376	00:44:32-01:26:19	880s	0.23	0	1.00
same UTC window on 13 other nights: 5 detections in 3.2 h (1.57/h, i.e. 0.38 expected)					
3250	01:00:26-01:44:08	848s	0.72	1	0.51
detection 01:04:10 UTC, 20-40 Hz, S = 33.76 dB					
3274	01:08:23-01:48:36	796s	0.16	0	1.00
3275	01:08:21-01:48:34	670s	0.12	0	1.00

Controls (published arrivals at 3315; D1's S near each, and how often background peaks are as strong):
05:03:01 3315: 2-5 Hz S=1.2 (160.29/h), 5-10 Hz S=10.2 (2.12/h), 10-20 Hz S=14.8 (1.89/h), 20-40 Hz S=17.3 (0.67/h)
05:18:13 3315: 2-5 Hz S=1.4 (145.88/h), 5-10 Hz S=3.6 (26.41/h), 10-20 Hz S=5.2 (30.19/h), 20-40 Hz S=8.9 (9.27/h)
01:33:44 3315: 5-10 Hz S=14.5 (1.2/h), 10-20 Hz S=13.8 (2.24/h), 20-40 Hz S=12.0 (2.59/h)

Exploratory controls, not pre-registered: T-phases of USGS M>=4 earthquakes (largest S per band near the prediction)
3315 2014-03-08T03:37:52 M5.2 187 km NE of Tual 3641 km: best 2-5 Hz S=3.7 at -6 s (9.43/h)
3250 2014-03-08T05:50:41 M4.0 199 km NE of Lospalo 1188 km: best 20-40 Hz S=20.2 at +21 s (15.08/h)
3250 2014-03-08T08:50:55 M4.4 233 km SSW of Pagar 2299 km: best 10-20 Hz S=20.8 at -36 s (3.22/h)
3274 2014-03-08T05:50:41 M4.0 199 km NE of Lospalo 3686 km: best 5-10 Hz S=5.1 at -88 s (10.78/h)
3275 2014-03-08T03:37:52 M5.2 187 km NE of Tual 3892 km: best 10-20 Hz S=4.4 at -24 s (14.96/h)
3275 2014-03-08T08:50:55 M4.4 233 km SSW of Pagar 5311 km: best 2-5 Hz S=9.9 at +58 s (7.95/h)